

Seminar: Image Segmentation Based on Visual Prompt

Group MNIST

VNUHCM - University of Science

December 16, 2025



23122008 - Mai Duc Minh Huy

23122024 - Chung Tin Dat

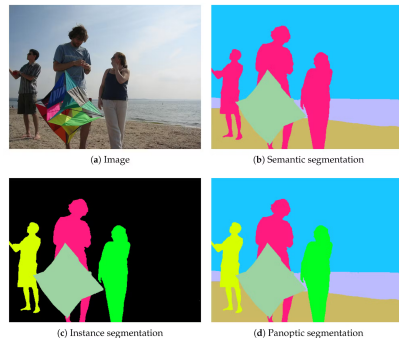
Table of contents

- 1 Introduction
 - General Background
 - Motivation
 - Problem Statement
 - Contribution
- 2 Related Works
 - Traditional Methods
 - Deep Learning Methods
- 3 Methodology
 - Data Collection
 - Model Architecture
- 4 Implementation & Experiment
- 5 Conclusion
- 6 Discussion
- 7 References



General Background

Image segmentation is a fundamental task in computer vision that involves partitioning a digital image into multiple segments or sets of pixels, often corresponding to different objects or parts of objects. The goal is to simplify or change the representation of an image into something that is more meaningful and easier to analyze.

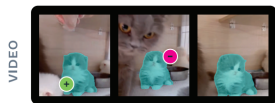
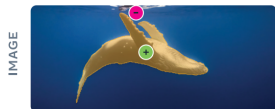


General Background

Visual prompts are user-provided graphical inputs—such as clicks, boxes, or lines that "tell" a computer vision algorithm where to focus or what specific object to process.

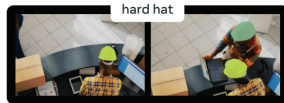
- **Points (Clicks):** The user clicks on a specific object (positive click) or background (negative click).
- **Bounding Boxes:** The user drags a rectangle around the object. This gives the algorithm a clear size and location limit.
- **Scribbles/Strokes:** The user draws a rough line over the object.

PROMPTABLE VISUAL SEGMENTATION



Prompts: positive  or negative  points

PROMPTABLE CONCEPT SEGMENTATION



Prompts: and/or positive  or negative  image exemplar

The history of this field can be broadly divided into two distinct eras.

- **Pre-Deep Learning Era (c. 1970s–2010s)**
- **Deep Learning Era (c. 2010s - Now)**



Pre-Deep Learning Era (c. 1970s-2010s)

Visual prompts function as constraints or "seeds." When a user marks a specific area, the algorithm treats those pixels as absolute truth—anchors that define the statistical profile (color intensity or texture) of the target object. The algorithm then propagates this information outwards to neighboring pixels, grouping them together based on how similar they are to the anchor points, stops only when it encounters a sharp change in contrast (an edge).



Pre-Deep Learning Era (c. 1970s-2010s)

Visual prompts function as constraints or "seeds." When a user marks a specific area, the algorithm treats those pixels as absolute truth—anchors that define the statistical profile (color intensity or texture) of the target object. The algorithm then propagates this information outwards to neighboring pixels, grouping them together based on how similar they are to the anchor points, stops only when it encounters a sharp change in contrast (an edge).

Prominent methods from this period include:

- Watershed
- Active Contour
- Grab Cut
- Region Growing



Pre-Deep Learning Era (c. 1970s-2010s)

Visual prompts function as constraints or "seeds." When a user marks a specific area, the algorithm treats those pixels as absolute truth—anchors that define the statistical profile (color intensity or texture) of the target object. The algorithm then propagates this information outwards to neighboring pixels, grouping them together based on how similar they are to the anchor points, stops only when it encounters a sharp change in contrast (an edge).

Prominent methods from this period include:

- Watershed
- Active Contour
- Grab Cut
- Region Growing

The prompt does not provide "understanding" of the object; it simply tells the math where to start calculating and what color values to look for nearby.



Deep Learning Era (c. 2010s)

Visual prompt is encoded into a high-dimensional vector and fused with the image features to act as a spatial query. Instead of just looking at neighbor pixel colors, the neural network uses the prompt to focus its "attention" mechanism on a specific region, combining the user's input with its own pre-trained knowledge of object shapes. For example, if a user clicks on a cat's ear, the model doesn't just select the ear's color; it recognizes the semantic pattern of a "cat" and predicts the entire animal.



Deep Learning Era (c. 2010s)

Visual prompt is encoded into a high-dimensional vector and fused with the image features to act as a spatial query. Instead of just looking at neighbor pixel colors, the neural network uses the prompt to focus its "attention" mechanism on a specific region, combining the user's input with its own pre-trained knowledge of object shapes. For example, if a user clicks on a cat's ear, the model doesn't just select the ear's color; it recognizes the semantic pattern of a "cat" and predicts the entire animal.

Key milestones include:

- Box Sup
- Scribble Sup
- Deep Extreme Cut
- Segment Anything



The drive to advance image segmentation stems from both scientific inquiry and practical necessity.

Scientifically,

- Segmentation addresses a fundamental question in perception: while image classification tells us what is in an image, segmentation answers where it is, down to the pixel level; segmentation helps reveal how an image is organized by separating it into meaningful regions.
- Many methods in computer vision rely on segmented regions as inputs. Examples include object detection, recognition, tracking, 3D reconstruction, and medical image analysis. Better segmentation improves the performance of these tasks.

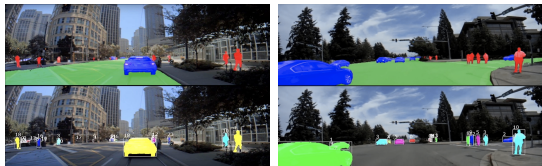


Practically, segmentation is an enabling technology for a multitude of real-world applications.

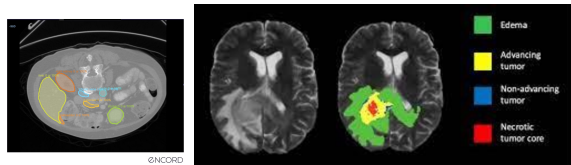


Motivation

Autonomous Driving



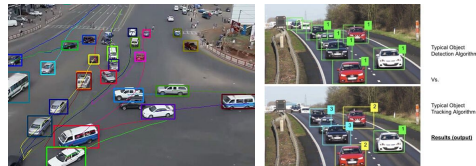
Medical Imaging



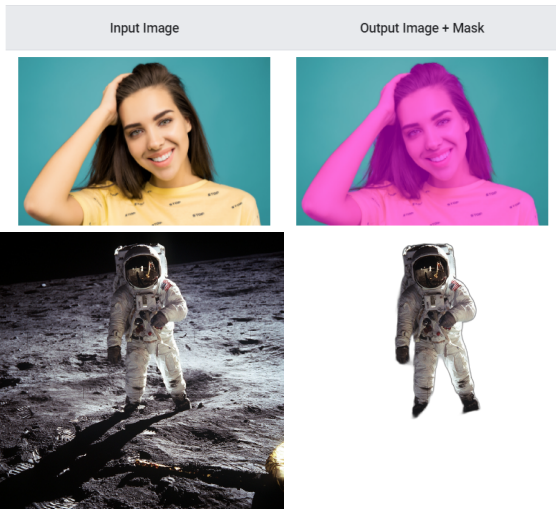
Surveillance



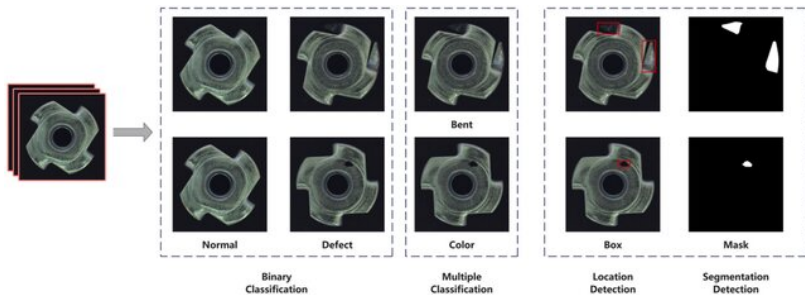
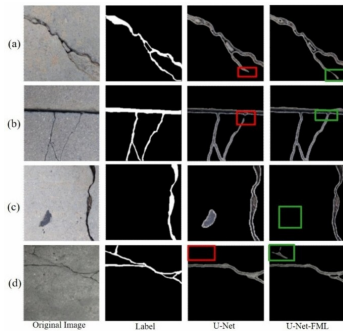
Object Tracking



Digital Photograph



Industrial Inspection



Segment Anything Model (SAM)

“In this work, our goal is to build a foundation model for image segmentation.” [?]

With this model, we aim to solve a range of downstream segmentation problems on new data distributions using prompt engineering.



Problem Statement

Image segmentation is the task of grouping pixels that belong to a common semantic object and separating them from other objects and the background.

The core principle of segmentation is guided by two criteria: pixels within the same group should be as similar as possible, while pixels belonging to different groups should be as distinct as possible.

In essence, the task simulates the human cognitive ability to transform raw, unstructured visual data into a structured representation of distinct entities.



Problem Statement

Let $\Omega \subset \mathbb{R}^2$ be the image domain and $I : \Omega \rightarrow \mathbb{R}^k$ the image.

A segmentation is a partition

$$\Omega = \cup_{i=1}^K \Omega_i \quad \Omega_i \cap \Omega_j = \emptyset \text{ for } i \neq j$$

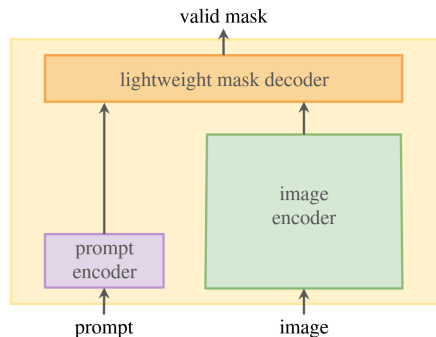
such that pixels within each region share similar properties according to some criterion.



Problem Statement

For a modern promptable segmentation system, the problem can be formally defined by its inputs and outputs.

- **Input:** An image and a "visual prompt" that specifies the object of interest. This prompt can take various forms, such as a point (or multiple points) on the object, a bounding box enclosing it, or even a rough mask.
- **Output:** One or more segmentation masks for the object(s) indicated by the prompt, along with a confidence score for each mask.



"Our principal contributions are a new task (promptable segmentation), foundational model for that new task (SAM), and dataset (SA-1B)"

A key contribution is the SA-1B dataset, which remains the largest segmentation dataset to date. It contains over 11 million high-resolution images and 1.1 billion high-quality segmentation masks.



The contribution of this report is to provide a comprehensive survey and structured analysis of the field of image segmentation. We synthesize the historical evolution, explain the motivations driving modern research, and offer a detailed examination of the methodology behind foundation models like SAM, contextualizing their impact and outlining future directions.

Traditional Methods

Methods	Mechanics
Watershed	Flooding process on gradient topography with controlled basins.
Active Contour	Energy minimizing curve driven by internal smoothness and image forces.
Grab Cut	Local similarity expansion based on pixel intensity or color distance.
Region Growing	Probabilistic graph cut with iterative foreground background modeling.



Traditional Methods

Methods	Methodology
	Stage 1: Input preparation
Watershed	Image gradient, foreground and background markers.
Active Contour	Image as continuous energy field, initial contour.
Grab Cut	Image as graph with color distributions (GMMs), bounding box.
Region Growing	Image as pixel grid with feature vectors, seed pixels.



Traditional Methods

Methods	Methodology	
	Stage 2: Propagation/Optimization	Stage 3: Final labeling
Watershed	Flooding until basins meet.	Watershed lines form region boundaries.
Active Contour	Iterative energy minimization via curve evolution.	Final curve represents object boundary.
Grab Cut	Repeated min cut optimization with model updates.	Cut defines segmentation boundary.
Region Growing	Iterative neighbor expansion.	Region boundary from grown region.



Traditional Methods

Methods	Contribution
Watershed	Introduced topographic interpretation of images and marker controlled segmentation to reduce over segmentation.
Active Contour	Formalized segmentation as energy minimization over curves and influenced decades of variational methods.
Grab Cut	Demonstrated practical interactive segmentation with probabilistic modeling.
Region Growing	Established seed based interactive segmentation.



Traditional Methods

Methods	Limitations
Watershed	Sensitive to noise and gradient quality.
Active Contour	Struggles with concave boundaries and weak edges.
Grab Cut	Assumes color separability between foreground and background; fails when distributions overlap.
Region Growing	Highly sensitive to noise and intensity variation, leakage occurs at weak boundaries.



Technique	Mechanics
Box Sup	Unsupervised methods to generates hundreds of potential mask shapes (candidate segments), bounding box is used to filter these candidates.
Scribble Sup	Image is divided into super pixels, use Graph Cuts optimization to label the unmarked pixels.
Deep Extreme Cut	Heatmap from extreme points, CNN extracts feature specifically on the areas highlighted by the heatmap.
Segment Anything	Cross attention between image embedding and prompt embedding.



Technique	Methodology	
	Image representation	Prompt encoding
Box Sup	CNN feature maps extracted from region proposals and full images.	Bounding boxes converted into soft foreground background supervision using objectness and low level segmentation cues.
Scribble Sup	CNN feature maps over full image.	Scribbles transformed into dense label maps via graph based or CRF propagation.
Deep Extreme Cut	CNN encodes image appearance along with spatial context.	Extreme points encoded as distance maps concatenated with image channels.
Segment Anything	Vision transformer generates dense image embeddings.	Prompts encoded into tokens representing points, boxes, or masks

Methods	Methodology
	Mask decoding
Box Sup	CNN trained on pseudo masks predicts dense semantic segmentation masks.
Scribble Sup	CNN outputs semantic masks learned from propagated pseudo labels.
Deep Extreme Cut	Single pass CNN inference produces object mask.
Segment Anything	Transformer based mask decoder predicts multiple candidate masks with confidence scores.



Technique	Contribution
Box Sup	.
Scribble Sup	.
Deep Extreme Cut	.
Segment Anything	First truly general purpose segmentation foundation model. Enables interactive segmentation, automatic segmentation, and box guided segmentation in one model.



Technique	Limitations
Box Sup	.
Scribble Sup	.
Deep Extreme Cut	.
Segment Anything	High computational cost during preprocessing due to large encoder. Not ideal for pixel perfect tasks such as thin structures or high precision medical boundaries without fine tuning. Mask quality depends on prompt placement.

The fundamental challenge in building a foundation model for segmentation is the lack of web-scale segmentation data.

Segmentation masks are not naturally occurring and must be manually created.

Traditional approaches would require paying annotators to label billions of masks-economically infeasible and temporally prohibitive.



SAM Data Engine:

- ① Assisted Manual Annotation (4.3M Masks from 120K Images)
- ② Semi-Automatic Annotation (5.9M Additional Masks, 180K Images)
- ③ Fully Automatic Annotation (1.1B Masks, 11M Images)



1. Assisted Manual Annotation (4.3M Masks from 120K Images):

Workflow:

- ① Professional annotators view an image in a web-based interface.
- ② They click foreground/background points to indicate an object of interest.
- ③ SAM's pre-computed image embedding (from MAE-ViT-H) is queried with these prompts.
- ④ The lightweight Prompt Encoder and Mask Decoder run in real-time ($< 50\text{ms}$), generating an initial mask prediction.
- ⑤ Annotators refine the mask using pixel-precise brush and eraser tools.
- ⑥ The final mask is saved.



Quantitative Dynamics:

- **Initial annotation time:** 34 seconds per mask (comparable to COCO annotation standards).
- **Model capacity progression:** ViT-B $\rightarrow$ ViT-H during this stage (1 retraining cycle per major improvement).
- **Average masks per image:** 20 $\rightarrow$ 44 (increased diversity as model improved).
- **Total masks collected:** 4.3M from 120K images.
- **Retraining cycles:** 6 full model retraining iterations.



Efficiency Gain Analysis:

The $34 \rightarrow 14$ second progression represents a **$6.5\times$ speedup over COCO standards** (34 seconds here vs. 14 seconds for bounding boxes). This exponential improvement in annotation velocity is crucial for economic feasibility at billion-scale.



2. Semi-Automatic Annotation (5.9M Additional Masks, 180K Images):

Workflow:

- ① Train a generic object detector on all masks from Stage 1. This detector outputs bounding boxes regardless of semantic class (object-agnostic proposal generation).
- ② SAM automatically segments high-confidence regions identified by this detector, generating candidate masks.
- ③ Annotators are presented with images pre-filled with these automatic masks.
- ④ Annotators' task: Label only the *unannotated* objects (those not covered by the automatic masks).
- ⑤ This forces annotators to focus on smaller, more occluded, or semantically ambiguous objects.



Quantitative Dynamics:

- **Annotation time (excluding automatic masks):** ~ 34 seconds per mask (unchanged; challenging objects require the same effort).
- **Average masks per image:** $44 \rightarrow 72$ (including automatic masks; diversity increased substantially).
- **New masks collected:** 5.9M, raising the cumulative total to 10.2M masks.
- **Retraining cycles:** 5 full model retraining iterations.



Efficiency Gain Analysis:

The $34 \rightarrow 14$ second progression represents a **$6.5\times$ speedup over COCO standards** (34 seconds here vs. 14 seconds for bounding boxes). This exponential improvement in annotation velocity is crucial for economic feasibility at billion-scale.



Why stage 2 is needed?

By automatically handling easy cases and forcing annotators to label hard cases, the model learns a more balanced representation. This is empirically validated: the model's ability to handle edge cases improved significantly from Stage 1 to Stage 2, as evidenced by the diversity metrics.



3. Fully Automatic Annotation (1.1B Masks, 11M Images):

Systematically prompt the model with a regular spatial grid of points (32×32), covering the entire image. Workflow:

- ① Each point is passed as a prompt to the Prompt Encoder.
- ② For each point, SAM predicts **three masks simultaneously**.
- ③ **Theoretical maximum:** $1024 \text{ points} \times 3 \text{ masks per point} = 3072 \text{ masks per image}$.



Why stage 3 is needed?

A regular grid ensures comprehensive spatial coverage. If an object's boundary passes through or near a grid point, the model will capture it from that point. Objects of varying sizes are covered because the grid is at a fixed, moderate resolution. This is fundamentally different from random point sampling, which can miss objects entirely.



Loss Function

A linear combination of focal loss and dice loss in a 20:1 ratio of focal loss to dice loss.

- Local Loss: Focus to learning energy to hard pixels.

$$FocalLoss = -\sum_{i=1}^n (i - p_i)^{\gamma} \log_b(p_i)$$

- Dice Loss: Differentiable approximation of IoU.

$$DiceLoss = \frac{2 * Intersection}{Prediction + Ground_Truth}$$

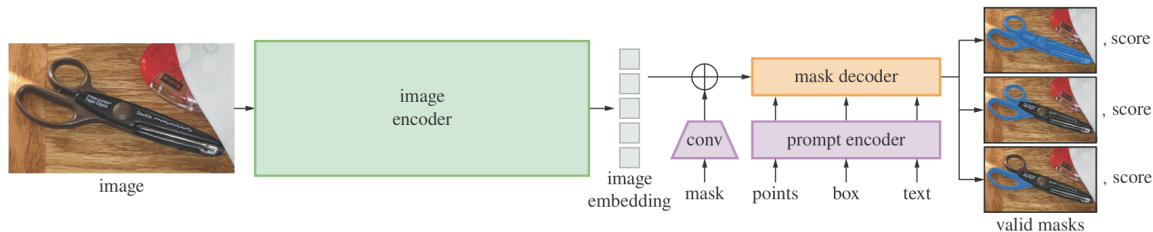


Why 3 outputs per prompt?

Empirical analysis during SAM's development showed that image regions typically have at most three levels of semantic hierarchy (whole > part > sub-part). Predicting three masks covers all common ambiguity scenarios without excessive computation.



Segment Anything Model

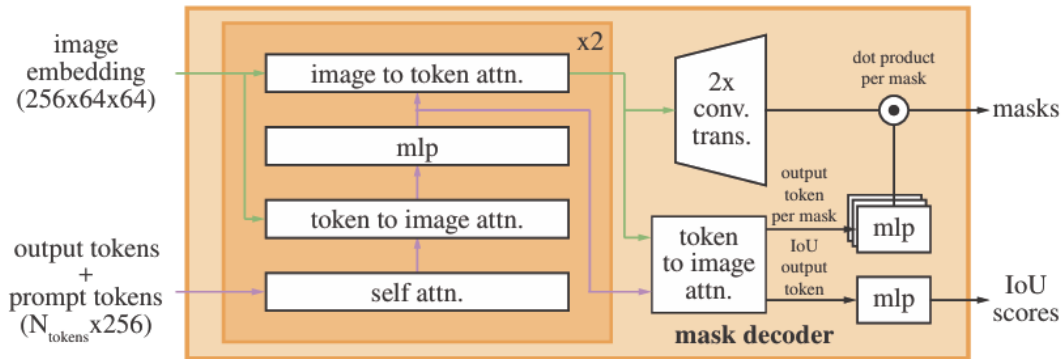


- Image Encoder: MAE pre-trained Vision Transformer (ViT).
- Prompt Encoder:
 - Point, box: Positional encoding summed with learned embeddings for each prompt type.
 - Mask: Convolution and summed element-wise with the image embedding.



Segment Anything Model

- Mask Decoder:



Segment Anything Model

Output token is an embedding vector acts as a query for the final mask prediction (analogous to CLS token in BERT or ViT).

This token gather information about the prompt embeddings and image embeddings to form semantic information required to segment the target object.



Segment Anything Model

- Image Encoder: Fine-tune its weights to adapt its visual features to be queryable.
- Mask Decoder: Transformer layers, MLP, upscaling convolutions.
- Prompt Encoder: Convolutional layers for positional encodings, vectors for "Foreground", "Background", "Top-left", "Bottom-right", "Not-a-point".
- Output token: 4 vectors (3 for multi-mask outputs: whole, part, sub-part, 1 for IoU confidence prediction).



References I



Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A. L., and Zhou, Y. (2021).

Transunet: Transformers make strong encoders for medical image segmentation.
arXiv preprint arXiv:2102.04306.



Comaniciu, D. and Meer, P. (2002).

Mean shift: a robust approach toward feature space analysis.
IEEE Transactions on Pattern Analysis and Machine Intelligence, 24(5):603–619.



Hartigan, J. A. and Wong, M. A. (2018).

A k-means clustering algorithm.
Journal of the Royal Statistical Society Series C: Applied Statistics, 28(1):100–108.



He, K., Gkioxari, G., Dollár, P., and Girshick, R. (2017).

Mask r-cnn.

In *Proceedings of the IEEE international conference on computer vision*, pages 2961–2969.



Nameirakpam, D., Singh, K., and Chanu, J. (2015).

Image segmentation using k -means clustering algorithm and subtractive clustering algorithm.

Procedia Computer Science, 54:764–771.